Where Was the Model Looking? Cheap Claim-Level Hallucination Detection from Attention and Token Probabilities

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Abstract

Claim-level hallucination detection for long-form LLM output is far from solved: on the MUCH benchmark the strongest published training-free method, Claim-Conditioned Probability (CCP), reaches 0.77 ROC-AUC while spending more compute than generation itself. We study GRASP, a detector that reads three interpretable attention statistics (grounding to the prompt, attention-sink leakage and dispersion) from a few label-free-selected heads per layer, pools them over relative depth, adds the token probabilities the model already produced, and scores each claim with a small gradient-boosted classifier. On MUCH's human-annotated test split GRASP reaches 0.924 ROC-AUC against CCP's 0.770 (Δ +0.154, 95% CI +0.110 to +0.199), wins on each of four open models, and costs 33–38% of generation time on a T4, where CCP measured on the same hardware costs 192–430%. Retrained with the same frozen recipe on RAGTruth it again beats CCP (0.827 vs 0.623). We report the limits as plainly: the detector is supervised; trained on MUCH and applied unchanged to RAGTruth it only ties CCP, even after label-free domain adaptation, and there the authors' pre-trained UHeads transfer better (0.680 vs 0.639); its training-free variants do not beat CCP. With in-domain labels GRASP also outperforms pre-trained UHeads on both benchmarks.

1 Introduction

An LLM states invented facts as fluently as true ones. For long answers, a single answer-level confidence score does not tell a user *which* sentence to check, so the useful unit is the atomic claim (Min et al., 2023). The MUCH benchmark (Dentan et al., 2026) makes this setting measurable: 4,873 generations from four open models in four languages, pre-segmented into 20,751 claims, with the generated tokens and their top-24 probabilities released. Its strongest baseline, CCP (Fadeeva et al., 2024), filters token-level uncertainty through a natural-language inference model applied at every token; it reaches 0.772 ROC-AUC at a cost the benchmark authors put above that of generation.

We ask whether the model's own attention, read in one teacher-forced pass, can do better and cheaper. The intuition is that when a model states a fact it holds, some heads retrieve from the prompt; when it improvises, attention drifts to its own recent output or collapses onto the attention sink. We make three contributions:

- A lightweight claim-level detector, GRASP, built from three interpretable attention statistics on label-free-selected heads plus free token probabilities, and a small tree model.
- An evaluation that follows MUCH's protocol exactly, keeps every modelling choice on the training split, and reports per model, across models, on a second benchmark (RAGTruth; Niu et al., 2024) and with runtime measured against CCP on the same GPU.

- An explicit account of what does *not* work: zero-shot cross-benchmark transfer and training-free variants.

2 Related work

Probability-based. Token likelihood and entropy are cheap but conflate lexical and factual uncertainty; CCP (Fadeeva et al., 2024) removes lexical variation with NLI at a large cost. **Sampling-based** methods such as SelfCheckGPT (Manakul et al., 2023) and semantic entropy (Kuhn et al., 2023) multiply generation cost and miss consistent errors. **Internal-state probes** predict hallucination from hidden states (Azaria and Mitchell, 2023; Chen et al., 2024) or attention: Lookback Lens (Chuang et al., 2024) uses the ratio of attention to context versus generated text, SinkProbe (Binkowski et al., 2026) the attention-sink mass, and RAUQ (Vazhentsev et al., 2026) selects uncertainty-aware heads without labels at sequence level. Closest to us, UHeads (Shelmanov et al., 2025) train a transformer head on attention features and token probabilities for claim-level detection. GRASP is in the same family; it differs in using three interpretable statistics from a few label-free-selected heads and a small tree model, and in being evaluated head-to-head against CCP on MUCH.

3 Method

3.1 Per-token attention signals

The generated tokens are replayed through the model once. For generated token t and a kept head, with attention row a_t over positions $0..t$ (position 0 is the sink, $1..P{-}1$ the prompt):

$$G_t = \frac{\sum_{j=1}^{P-1} a_{t,j}}{\sum_{j\geq 1} a_{t,j}} \qquad \text{(grounding to the prompt)} \tag{1}$$

$$S_t = a_{t,0} \qquad \text{(sink leakage)} \tag{2}$$

$$D_t = \frac{-\sum_{j\geq 1} \tilde{a}_{t,j} \log \tilde{a}_{t,j}}{\log |\{j \geq 1\}|} \qquad \text{(dispersion)} \tag{3}$$

where $\tilde{a}$ is the non-sink row renormalised. The row used is the one that *produced* token t. A saliency weight W_t is the attention token t receives from later generated tokens, divided by their number.

3.2 Label-free head selection

For each layer we keep the k heads whose G varies most across tokens on 200 unlabelled generations of the same model; a head with near-constant G cannot discriminate. We use k=4.

3.3 Claim features and classifier

Layers are pooled into 8 relative-depth bins so that models of different depth share one feature space. For each bin, kept head and signal, a claim's content tokens (stopwords and punctuation removed) give the mean, max, min and saliency-weighted mean of the per-(layer, head) z-scored value (384 features). We add seven token-probability features from the top-24 probabilities MUCH already releases (chosen-token surprisal, entropy, top-1 probability, top-1/top-2 gap) and two claim-length features. A gradient-boosted tree classifier (31 leaves, 800 rounds) predicts the hallucination label. **GRASP-free**, a training-free variant, averages the per-model rank of mid-depth minimum grounding with the rank of token surprisal and entropy, with signs fixed a priori.

4 Experimental setup

Data. MUCH, all 8 generation configs (Llama-3.2-3B, Llama-3.1-8B, Gemma-3-4B and Ministral-8B, at temperatures 0.7 and 1.0). MUCH's test split is its 200 human-annotated generations (867 claims); the train split is the other 4,673 (19,884 claims). We follow MUCH's evaluator: GPT-4o labels on claims where GPT-4o and GPT-4.1 agree, the trailing end-of-sequence chunk dropped, and stopwords masked for probability baselines.

Protocol. Feature groups, normalisation, learner and k were chosen by 5-fold cross-validation on the train split, grouped by generation and ranked by the *mean of per-model* ROC-AUCs, since pooled ROC can be inflated by recognising which model wrote a claim (hallucination rates range from 22% to 54%). The selection was frozen to a file before the test split was read. This was the second reading of MUCH's test split; the first evaluated an earlier version (§6).

Baselines. CCP as released by MUCH (δ=24, σ=8, `product` aggregation), which we reproduce at 0.775 ROC / 0.643 PR (published: 0.772 / 0.639); token surprisal and entropy; and UHeads (Shelmanov et al., 2025) via the authors' pre-trained claim heads (§5.4).

Statistics. 95% CIs come from 2,000 bootstrap resamples of generations; differences are paired on identical claims.

5 Results

5.1 MUCH

Method	Labels	ROC-AUC [95% CI]	PR-AUC	TPR@10%	ΔROC vs CCP
CCP	no	0.770 [0.724, 0.817]	0.637	0.537	–
GRASP-0 (hand-weighted)	no	0.629 [0.569, 0.687]	0.388	0.188	−0.141
GRASP-free	no	0.764 [0.713, 0.815]	0.501	0.354	−0.006 [−0.056, +0.041]
GRASP	yes	**0.924** [0.896, 0.948]	**0.845**	**0.790**	**+0.154** [+0.110, +0.199]
GRASP + CCP input	yes	0.920 [0.895, 0.944]	0.838	0.729	+0.150 [+0.111, +0.191]

Table 1: MUCH human-annotated test split (867 claims, 229 hallucinated).

GRASP beats CCP on every model: Gemma-3-4B 0.917 vs 0.726, Llama-3.1-8B 0.937 vs 0.831, Llama-3.2-3B 0.889 vs 0.767 and Ministral-8B 0.897 vs 0.830; all paired CIs exclude zero. Adding CCP as an input does not help, so its NLI step is unnecessary. At response level (does an answer contain any hallucination; 200 generations) GRASP reaches 0.911 vs CCP's 0.762 (Δ +0.149, CI +0.086 to +0.214).

Transfer to an unseen model. Trained on three models' training claims and scored on the fourth model's test claims, GRASP beats CCP on Gemma (0.808 vs 0.726), Llama-3.1-8B (0.882 vs 0.831) and Ministral (0.853 vs 0.830) and trails it on Llama-3.2-3B (0.759 vs 0.767).

What contributes. In train cross-validation (mean per-model ROC; CCP 0.797), token probabilities alone give 0.771, attention alone 0.832, both 0.866, and the tuned learner 0.885. Four heads per layer beat one consistently (+0.01 to +0.02).

5.2 Abstention

Withholding the claims GRASP rates riskiest lowers the hallucination rate of what remains from 26.4% to 11.4% at 20% withheld and to 7.2% at 30% (CCP: 16.3% and 14.7%); the area under the risk–coverage curve is 0.062 vs 0.152.

5.3 Second benchmark: RAGTruth

We apply the frozen recipe to RAGTruth's Mistral-7B-Instruct responses (retrieval-augmented QA, summarisation, data-to-text). Claims come from MUCH's segmenter and are hallucinated if they overlap an annotated span; the test set has 445 responses and 17,226 claims (7.7% hallucinated). Rules were fixed before extraction (Table 2). Retrained in-domain, GRASP wins on every task (QA

Method	Trained on	ROC-AUC [95% CI]	PR-AUC	ΔROC vs C
CCP	–	0.623 [0.602, 0.644]	0.150	–
mean token entropy	–	0.623	0.114	−0.000
GRASP-free	–	0.595	0.098	−0.028 [−0.057,
GRASP, zero-shot	MUCH	0.625 [0.606, 0.645]	0.118	+0.002 [−0.025,
GRASP, zero-shot + rank norm.	MUCH + unlabelled RAGTruth	0.639 [0.617, 0.659]	0.131	+0.015 [−0.011,
GRASP	RAGTruth train	**0.827** [0.803, 0.849]	**0.383**	**+0.203** [+0.177,

Table 2: RAGTruth test, claim level.

0.859, summary 0.765, data-to-text 0.853, vs CCP 0.626, 0.617, 0.624). Zero-shot from MUCH it only ties CCP: better on QA and summaries, worse on data-to-text. As a label-free remedy we compared, by MUCH leave-one-model-out on the training split, the frozen recipe (mean held-out ROC 0.808), per-domain rank normalisation of every feature fitted on unlabelled target features (0.819) and rank normalisation with importance weighting (0.818). The selected rank normalisation lifts zero-shot RAGTruth ROC to 0.639, still not significantly above CCP.

5.4 Comparison with UHeads

We ran the authors' pre-trained claim-level UHeads (Shelmanov et al., 2025), teacher-forced on exactly the claims GRASP is scored on. Our reimplementation of their attention features was checked equal to their extractor. The score's sign and, on RAGTruth, the choice between their two Mistral heads were fixed on training claims only. With in-domain labels GRASP outperforms

Benchmark	Method (labels)	ROC-AUC [95% CI]	ΔROC vs UHead
MUCH, Llama-3.1-8B (282 claims)	UHead (pre-trained)	0.570 [0.486, 0.653]	–
	CCP	0.831	+0.261
	GRASP (other three models)	0.883 [0.817, 0.938]	+0.313 [+0.234, +0.397]
	GRASP (in-domain)	**0.937**	+0.367 [+0.289, +0.449]
RAGTruth, Mistral (17,181 claims)	CCP	0.623	−0.057
	GRASP zero-shot + rank norm. (MUCH)	0.639	−0.041 [−0.076, −0.004]
	UHead (pre-trained)	0.680 [0.649, 0.708]	–
	GRASP (RAGTruth train)	**0.827**	+0.147 [+0.116, +0.180]

Table 3: GRASP vs pre-trained UHeads on identical claims.

UHeads on both benchmarks. Zero-shot, UHeads' pre-trained Mistral head transfers to RAGTruth better than GRASP trained on MUCH alone; it was trained on far larger and more varied data. The pre-trained Llama head is weak on MUCH even on training claims (oriented ROC 0.526), while the same code is sensible on RAGTruth. We therefore attribute this to a mismatch between its training claims and MUCH's very short claims (about 4.6 tokens), and treat the MUCH margin over UHeads as indicative only.

5.5 Runtime

On Tesla T4s, GRASP's extraction pass costs 33–38% of measured generation time. MUCH's CCP, timed on the same hardware, costs 235–430% on MUCH's models and 192% on RAGTruth-length answers. CCP's NLI cost is fixed per token, so it is relatively worst for small, fast models.

6 Limitations

GRASP is supervised and needs labels from the setting it monitors; without them it transfers across models but not across benchmarks and task families. Training-free variants do not beat CCP at claim level on either benchmark, and at response level the RAGTruth advantage of the training-free score (0.679 vs 0.647) does not hold on MUCH (0.616 vs 0.762). Without in-domain labels, pre-trained UHeads transfer better than GRASP (RAGTruth 0.680 vs 0.639), and our comparison uses their released heads rather than retraining them on MUCH. MUCH's test split was read twice: once by an earlier version and once by GRASP as reported, with all choices frozen on the training split beforehand. Test labels are GPT-4o's, applied to the human-annotated samples, as in MUCH's evaluator. CCP and generation were timed on different sample generations, so overheads compare per token. RAGTruth responses were replayed through Mistral-7B-Instruct-v0.1, assumed to be the generator version.

7 Conclusion

Reading where the model looked, through a few interpretable attention statistics, plus the probabilities it already produced, gives a claim-level hallucination detector that clearly beats CCP and pre-trained UHeads on MUCH and, retrained, on RAGTruth, at a fraction of CCP's cost. The open problem it leaves is transfer to new task families without labels, where large pre-trained heads currently do better.

References

Amos Azaria and Tom Mitchell. The internal state of an llm knows when it's lying, 2023. URL https://arxiv.org/abs/2304.13734.

Jakub Binkowski, Kamil Adamczewski, and Tomasz Kajdanowicz. Attention sinks as internal signals for hallucination detection in large language models, 2026. URL https://arxiv.org/abs/2604.10697.

Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. Inside: Llms' internal states retain the power of hallucination detection, 2024. URL https://arxiv.org/abs/2402.03744.

Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James Glass. Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps, 2024. URL https://arxiv.org/abs/2407.07071.

Jérémie Dentan, Alexi Canesse, Davide Buscaldi, Aymen Shabou, and Sonia Vanier. Much: A multilingual claim hallucination benchmark, 2026. URL https://arxiv.org/abs/2511.17081.

Ekaterina Fadeeva, Aleksandr Rubashevskii, Artem Shelmanov, Sergey Petrakov, Haonan Li, Hamdy Mubarak, Evgenii Tsymbalov, Gleb Kuzmin, Alexander Panchenko, Timothy Baldwin, Preslav Nakov, and Maxim Panov. Fact-checking the output of large language models via token-level uncertainty quantification. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 9367–9385, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.558. URL https://aclanthology.org/2024.findings-acl.558/.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, 2023. URL https://arxiv.org/abs/2302.09664.

Potsawee Manakul, Adian Liusie, and Mark Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.557. URL https://aclanthology.org/2023.emnlp-main.557/.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.741. URL https://aclanthology.org/2023.emnlp-main.741/.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Artem Shelmanov, Ekaterina Fadeeva, Akim Tsvigun, Ivan Tsvigun, Zhuohan Xie, Igor Kiselev, Nico Daheim, Caiqi Zhang, Artem Vazhentsev, Mrinmaya Sachan, Preslav Nakov, and Timothy Baldwin. A head to predict and a head to question: Pre-trained uncertainty quantification heads for hallucination detection in LLM outputs. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 35712–35731, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1809. URL https://aclanthology.org/2025.emnlp-main.1809/.

Artem Vazhentsev, Lyudmila Rvanova, Gleb Kuzmin, Ekaterina Fadeeva, Ivan Lazichny, Alexander Panchenko, Maxim Panov, Mrinmaya Sachan, Preslav Nakov, Timothy Baldwin, and Artem Shelmanov. Efficient hallucination detection for llms using uncertainty-aware attention heads, 2026. URL https://arxiv.org/abs/2505.20045.